

ECE422 MIDTERM — FORMULAS & KEY FACTS (front)

Professor's conventions: Manchester **0 = low** → **high**, **1 = high** → **low** (= NRZ XOR CLK, draw CLK line) · 2-D parity corner = parity of the **P column** (HW2: P col 0,0,1 → corner 0) · checksum: add, add, **plus carry**, complement · count errors with $n!/(k!(n-k)!)$ · CRC R may be written with a leading 0 (EX3: 0111).

CALCULATIONS

Topic	Rule	Example
Shannon	$C = B \cdot \log_2(1+S/N)$; dB = $10 \log_{10}(S/N)$ (30 dB → 1000)	$5000 \cdot \log_2 16 = 20,000$ b/s
$\log_2$ trick	$2^{10}=1024 \approx 1000$; $\log_2 x = \log_{10} x / 0.301$	$\log_2 11 \approx 3.46$
Min frame L	$L = 2 \cdot d \cdot R / c$	$2 \cdot 2500 \cdot 10^7 / 2 \cdot 10^8 = 250$ bits
Ethernet min	51.2 μ s round trip × 10 Mb/s	512 bits = 64 B
Backoff	n collisions → $k \in 0 \dots 2^n - 1$, wait $k \cdot 51.2 \mu$ s	$n=3 \rightarrow \max 7 \cdot 51.2 = 358.4 \mu$ s
Wavelength	$\lambda = \text{speed}/f$; copper $\approx 2/3 \cdot 3 \cdot 10^8$	300 Hz copper → 667 km
Overhead	headers/(headers+data)	$4 \times 20B, 1B \rightarrow 80/81$
Errors	$C(n,k)$; n includes P	$C(5,2) = 10$
MAC → dec	read hex as one number	0A:00 = 0x0A00 = 2560

ENCODING

- NRZ** 1 = high, 0 = low. Problems: **baseline wander**, **clock recovery**.
- NRZI** 1 = transition, 0 = stay (fixes 1s, not 0s). Start level only matters for NRZI.
- Manchester** mid-bit transition, bit rate = $\frac{1}{2}$ baud rate (10 Mb/s → 20 Mbaud, 50%). Used by Ethernet.
- 4B/5B** ≤ 1 leading 0, ≤ 2 trailing 0 → ≤ 3 zeros in a row, 80%, sent with NRZI. 0000 → 11110, 0001 → 01001, 0010 → 10100, 1111 → 11101; 11111 idle, 00000 dead, 00100 halt.
- 2B/3B** (EX3): no > 2 zeros → drop codes starting or ending with 00 → 010, 011, 101, 110, 111.
- FSK**: same amplitude, one frequency for 1, another for 0.

PARITY

- Even: total 1s even; odd: total 1s odd. Detects **odd** # of errors, misses **every even** # (each flip ± 1).
- 1001P odd → P=1 → 10011; all $C(5,2)=10$ 2-bit errors missed: 01011 00111 00001 00010 11111 11001 11010 10101 10110 10000.
- 2-D: row P + column P byte; catches all 1/2/3-bit, most 4-bit (misses rectangle); corrects 1 bit (bad row × bad col). Used in BISYNC.
- 1-bit CRC must use $C(x)=x+1=11 \rightarrow$ **same as even parity** (XOR 11 changes # of 1s by 0 or 2).

HW2 odd 2-D: 00010101|0 10110011|0 01000001|1
col P 00011000 | corner 0 (from P col)

CHECKSUM (1'S COMPLEMENT)

00010101 + 10110011 = 11001000 (Sum1)
+ 01000001 = 1 00001001 (Sum2)
plus carry = 00001010
complement = 11110101 ← transmitted
Receiver: sum all + checksum = 11111111 → complement 0 = OK

- Carry can be "10" (3 words) — add it back as is. Internet checksum = 16-bit words, IP (not link level).
- Misses reordered fields and errors that cancel (same bit position flips 0→1 in one word, 1→0 in another).
- Hex fields: a2:06:23 → sum cb → checksum 34.

CRC

- Bits ↔ polynomial: $110001 = x^5 + x^4 + 1$. $C(x)$ degree $r \rightarrow$ append **r zeros**, XOR-divide, remainder (r bits) replaces the zeros. Receiver

remainder 0 = OK. Only $C(x)$ must be known.

- Window: if it starts with 1 → XOR $C(x)$; drop first bit; bring down next bit.
- Detects: all 1-bit (x^k and x^0 terms $\neq 0$); all 2-bit (factor with ≥ 3 terms); any odd # (factor $x+1$); bursts < k bits. Used in HDLC, DDCMP, CSMA/CD, Token Ring. CRC-32 for Ethernet.
- $C(x) = x^m$ (e.g. 100) → $R=0$ iff last m bits are 0 → 6-bit M: $2^4-1 = 15$ values (4, 8, ..., 60).

M=1101, C=1011: 1101000 → 110101011=0110 → 110001011=0111 → 111001011=0101 → 101001011=0001 → R=001 → send 1101001

ETHERNET 802.3

- Xerox PARC mid-70s; root = **ALOHA** (Hawaii); DEC+Intel+Xerox 10 Mb/s 1978; Fast 100, Gigabit 1000.
- CSMA/CD**, Manchester, **1-persistent**. Segment 500 m coax (50 Ω), ≤ 4 repeaters, reach 2500 m. 10Base2 = 10 Mb/s baseband 200 m; 10BaseT = twisted pair 100 m, hub.
- Frame: preamble 64 · dst 48 · src 48 · type 16 · data 46–1500 B · CRC 32. Min 64 B = 14 hdr + 46 data + 4 CRC.
- Collision → 32-bit jam; runt = 96 bits (64+32). Worst case detect at $t + 2d$.
- Address 48 bits, in adaptor ROM, 24-bit maker prefix (AMD 8:0:20). All 1s = broadcast; first bit 1 = multicast. Accepts own, broadcast, chosen multicast. < 200 hosts typical (max 1024).

CSMA / ALOHA

	Idle	Busy
ALOHA	send any time, no carrier sense	
1-persist.	send	wait until idle, send at once
Non-persist.	send	wait random time, retry
p-persist.	send w.p. p, defer slot w.p. 1-p	wait random time

Data link = **LLC + MAC** sublayers.

WIRELESS / 802.11

- FCC allocates; license-exempt → power limit + spread spectrum (**frequency hopping** / **direct sequence**, n -bit chipping → $n \times$ wider).
- 802.11: FH 79×1 MHz or DS 11-bit chip, 2.4 GHz 2 Mb/s · b 11 Mb/s 2.4 GHz · a 54 Mb/s OFDM 5 GHz · g 54 Mb/s OFDM 2.4 GHz.
- Hidden**: A, C reach B not each other → collide at B. **Exposed**: C hears B → A, wrongly won't send to D.
- MACA**: RTS → CTS (length). Hear CTS → quiet; hear only RTS → may send. MACAW adds ACK. No CD: no CTS → backoff.
- APs + distribution system (layer 2). Active scan: Probe → Probe Resp → Assoc Req → Assoc Resp. Passive: Beacon.
- Frame: 4 addr, data ≤ 2312 B, CRC 32. ToDS=FromDS=1: Addr1 final dst, Addr2 last AP, Addr3 first AP, Addr4 source (E, AP-3, AP-1, A).
- Bluetooth 2.45 GHz, 10 m, PAN, 2.1 Mb/s, piconet 1 master + 7 slaves (parked). ZigBee 802.15.4 low power.

LAYERS, INTERNET, HISTORY

- TCP/IP 4 layers: App, Transport, Network, Link; routers = bottom 2. OSI 7 (adds Phy, Session, Presentation).
- DL: frames over a link (sync, error, flow) · Net: routing, addressing · Transport: reliable delivery, flow, error recovery.
- HW (MAC) @ only to the next hop / closest router; same LAN = same network ID.
- Protocol = syntax, semantics, timing. Standard = agreed rules. Internet Draft (6 months) → RFC. IETF, IAB, ISOC, IRTF, IANA, ICANN.
- 1957 Sputnik → ARPA · 1962–68 packet switching · 1969 ARPANET (UCLA, SRI, UCSB, Utah) · 1971 e-mail · 1973 TCP/IP · 1982 TCP/IP for ARPANET · 1984 DNS · 1991 WWW (CERN) · 1993 Mosaic.

- 1 Pkt dst 3 hops away — dst HW@ after hop 1? → **Hop 2's HW@**
- 2 FSK of 10011? → **figure: one frequency for 1, another for 0**
- 3 $B = 5000 \text{ Hz}$, $S/N = 15$, max rate? → **20,000 b/s** ($5000 \cdot \log_2 16$)
- 4 2B/3B codes with no > 2 consecutive 0s? → **010, 011, 101, 110, 111**
- 5 $C(x) = 100$, 6-bit M: # with $R = 0$? → **15 values: 4, 8, ..., 60** (last 2 bits 00)
- 6 Sublayers of Data Link? → **LLC and MAC**
- 7 Protocol with no carrier sense? → **ALOHA**
- 8 Idle → wait random time: persistency? → **Non-persistent and p-persistent**
- 9 $d = 2500 \text{ m}$, $R = 10 \text{ Mb/s}$, $c = 2 \cdot 10^8$: min L? → **250 bits** ($2dR/c$)
- 10 Hidden station problem? → **Tx close to receiver but out of range of the other transmitter**
- 11 Layers in OSI not in TCP/IP? → **Physical, Session, Presentation**
- 12 Decimal of 00:00:00:00:0A:00? → **2560** ($10 \cdot 256$)
- 13 R if C = 1011, M = 11001010111001? → **R = 0111** (= 111, after +000)
- 14 Pkt to Instagram — how far does phone HW@ go? → **Up to your closest router**
- 15 How to know hosts are in the same LAN? → **Same network ID (masked net @)**
- 16 How is a packet network resilient to destruction? → **Metadata + packet routing/forwarding**
- 17 Internet killer application? → **E-mail, WWW**
- 18 4 early adopter institutions? → **U. Utah, Stanford R.I., UCLA, UCSB**
- 19 Why did Sputnik push communications? → **Tolerance to partial destruction**
- 20 Transport layer functions? → **Reliable delivery, flow control, error recovery**
- 21 CSMA/CD? → Carrier Sense Multiple Access with Collision Detection
- 22 Ethernet's root protocol? → ALOHA, University of Hawaii
- 23 Where/when Ethernet? IEEE std? → Xerox PARC, mid-1970s; 802.3
- 24 Max repeaters; total reach? → 4; 2500 m
- 25 10Base2? → 10 Mb/s, baseband, $\leq 200 \text{ m}$ segment
- 26 10BaseT? → twisted pair, 100 m, hub
- 27 Ethernet encoding? → Manchester
- 28 Ethernet frame fields? → preamble 64, dst 48, src 48, type 16 bits, data 46–1500 B, CRC 32
- 29 Min Ethernet frame and why? → 512 bits = 64 B, to detect collisions ($51.2 \mu\text{s}$ RTT)
- 30 Jam sequence; runt frame? → 32 bits; 96 bits (64 + 32)
- 31 Ethernet persistency? → 1-persistent
- 32 Exponential backoff? → $k \cdot 51.2 \mu\text{s}$, k random in $0 \dots 2^n - 1$
- 33 Ethernet address size/owner? → 48 bits; adaptor ROM
- 34 Broadcast? Multicast? → all 1s; first bit 1 (not all 1s)
- 35 Manufacturer prefix? → 24 bits (AMD 8:0:20)
- 36 p-persistent when idle? → send w.p. p, defer w.p. $1-p$
- 37 Spread spectrum techniques? → frequency hopping, direct sequence
- 38 License-exempt restrictions? → power limit + spread spectrum
- 39 Who allocates bands (USA)? → FCC
- 40 802.11b / a / g? → 11 Mb/s 2.4 GHz / 54 Mb/s 5 GHz / 54 Mb/s 2.4 GHz
- 41 Exposed node? → hears a transmission and wrongly holds back
- 42 RTS / CTS? → Request To Send / Clear To Send
- 43 Hear CTS? Hear only RTS? → don't send; free to send
- 44 How 802.11 detects collision? → no CTS → backoff (no CD)
- 45 Active vs passive scanning? → node sends Probe / AP sends Beacon
- 46 Frames to join an AP? → Probe, Probe Resp, Assoc Req, Assoc Resp
- 47 802.11 addresses; max data? → 4; 2312 B
- 48 Bluetooth band, range, network? → 2.45 GHz, 10 m, piconet (PAN)
- 49 Piconet size? → 1 master + 7 slaves
- 50 ZigBee? → IEEE 802.15.4, very low power, low bandwidth
- 51 Protocol elements? → syntax, semantics, timing
- 52 Internet Draft lifetime? → 6 months, then may become an RFC
- 53 Who publishes RFCs? → IETF
- 54 Routers implement? → bottom two layers (Network + Link)
- 55 First ARPANET message? → "LO", then crash (Oct 29, 1969)
- 56 TCP/IP invented? standard for ARPANET? → 1973; 1982
- 57 DNS introduced? WWW released? → 1984; 1991 (CERN, Berners-Lee)
- 58 NRZ problems? → baseline wander, clock recovery
- 59 NRZI encodes 1 as? → a transition
- 60 Manchester bit rate vs baud rate? → bit rate = $\frac{1}{2}$ baud rate
- 61 4B/5B efficiency? → 80%
- 62 Parity detects which errors? → odd numbers of bit errors
- 63 2-D parity catches? → all 1-, 2-, 3-bit, most 4-bit; corrects 1
- 64 2-D parity used in? Checksum in? → BISYNC; IP
- 65 CRC detects any odd # of errors if? → $C(x)$ has factor $(x+1)$
- 66 CRC: what must both sides know? → only $C(x)$
- 67 1-bit CRC vs parity? → same ($C(x) = 11$ = even parity)
- 68 BISYNC SOH / STX / ETX / DLE? → start header / start text / end text / escape
- 69 Connection-oriented vs connectionless? → telephone (setup, guaranteed) vs postal (each pkt routed)
- 70 Multiplexing types? → TDM, FDM, statistical